



Constellation

Clinton Power Station
8401 Power Road
Clinton IL 61727

U- 604858

10 CFR 50.73

December 16, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2025-003-00

Enclosed is Licensee Event Report (LER) 2025-003-00: SCRAM Turbine Hydraulic Oil Leak.
This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Derek Hillinger,
Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Frank Payne
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2025-003-00

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

IE22
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Clinton Power Station

☒ 050
☐ 052

2. Docket Number

00461

3. Page

1 OF 3

4. Title

Manual Scram Due to Main EHC Leak

5. Event Date

Month	Day	Year
10	22	25

6. LER Number

Year	Sequential Number	Revision No.
2025	- 003 -	00

7. Report Date

Month	Day	Year
12	22	25

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
N/A		

Facility Name	☐ 052	Docket Number
N/A		

9. Operating Mode

Mode 1

10. Power Level

86%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Derek Hillinger, Regulatory Assurance Manager

Phone Number (Include area code)

(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	TG	FCV	G080	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 22, 2025, at 1535, the Clinton Unit 1 reactor was manually scrambled due to low turbine Main Electro-Hydraulic Control oil system (Main EHC) reservoir level caused by a leak that had developed on Turbine Control Valve #3 (TCV3).

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) for "Any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph (a)(2)(iv)(B) of this section." The specific system listed in that paragraph that actuated was the "Reactor protection system (RPS) including: reactor scram or reactor trip."

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Clinton Power Station	☒ 050	2. DOCKET NUMBER 00461	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**CONDITION PRIOR TO THE EVENT**

Unit: 1
Mode: 1

Event Date: October 22, 2025,
Mode Name: Power Operation

Event Time: 15:35 CST
Reactor Power: 86%

EVENT DESCRIPTION

On October 22, 2025, 1504, operators observed abnormal valve conditions (main turbine bypass valves open and TCV3 indicated closed) and initiated off-normal procedures to lower power in response to unexpected loss of feedwater heating. Main EHC reservoir level was observed to be dropping rapidly (approximately 1 inch/minute), prompting an immediate operator action to makeup reservoir level. Due to the large leak being un-isolable and Main EHC fluid level reaching the low reservoir level in which Main EHC pumps are required to be secured, a manual scram was inserted at 1535 with the reactor at 86% power. Final Main EHC fluid reservoir level was -24.5" and an estimated 440 gallons of EHC fluid leaked during this event.

CAUSE OF THE EVENT

The apparent cause of the Main EHC leak was a failure of four tie rods associated with TCV3 due to high cyclic fatigue as a result of inadequate torquing during the most recent refurbishment by the vendor.

Post-SCRAM inspections identified significant mechanical damage to the TCV3 actuator, including a dislodged backing plate, and sheared four of the eight tie rods. Two of the remaining four tie rods were found to be under-torqued (one finger tight, the other at under 200 ft-lbs). The TCV3 actuator has since been replaced to correct the source of the leak. During the extent of condition investigation, Turbine Control Valve #1 (TCV1) actuator tie rods failed an initial resonance test and were replaced to preclude a similar failure. While preparing to install the replacement actuator for TCV1, the tie rods on the new actuator were torque-tested and three were found to be under torqued at 350 ft-lbs rather than the prescribed torque of 500 ft-lbs. In addition, the lower O-ring was visible in the groove when observed with a borescope from inside the cylinder. When properly torqued, the lower O-ring was no longer visible.

SAFETY CONSEQUENCES

There were no safety consequences impacting plant or public safety as a result of this event. The trip was not complex, with all systems responding normally post-trip. Decay heat was removed post-trip by discharging steam to the Main Condenser using the Turbine Bypass Valves which have a separate electro-hydraulic oil system. There was no loss of any safety function that would have prevented fulfillment of actions necessary to 1) shutdown the reactor and maintain it in a safe shutdown condition, 2) remove residual heat, 3) control the release of radioactive material, or 4) mitigate the consequences of an accident.



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NARRATIVE

CORRECTIVE ACTIONS

The station completed replacement of the TCV3 and TCV1 actuators, as well as revised the procurement requirements to require a Certificate of Conformance from the vendor regarding tie rod torquing requirements for these types of components. A Corrective Action Program Evaluation (CAPE) Investigation was conducted to further investigate the cause.

PREVIOUS OCCURRENCES

A review of Licensee Event Reports for the past five years identified no previous similar occurrences at Clinton Power Station.